

Question:

A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide $K_4Fe(CN)_6$. After filtering, the precipitate is dried and weighed. The reactions which occur are $ZnS + 2HCl \rightarrow ZnCl_2 + H_2S$ $2ZnCl_2 + K_4Fe(CN)_6 \rightarrow Zn_2Fe(CN)_6 + 4KCl$ If a 2 gram sample of rock yields 0.969 gram of $Zn_2Fe(CN)_6$, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight $Zn_2Fe(CN)_6$ = 342.6.

Answer Choices:

- A. 10 % Zn
- B. 22 % Zn
- C. 20 % Zn
- D. 19 % Zn
- E. 15 % Zn
- F. 18 % Zn
- G. 16 % Zn
- H. 14 % Zn
- I. 12 % Zn
- J. 25 % Zn

Reasoning + Answer:

<think>

- $ZnS + 2HCl \rightarrow ZnCl_2 + H_2S$: Zn is stoichiometrically present in ZnS and $ZnCl_2$.

- $ZnCl_2 + K_4Fe(CN)_6 \rightarrow Zn_2Fe(CN)_6 + 4KCl$: Zn is stoichiometrically present in $Zn_2Fe(CN)_6$.

- Therefore the amount of $Zn_2Fe(CN)_6$ produced is stoichiometrically proportional to Zn in ZnS.

- Molecular weight of $Zn_2Fe(CN)_6 = 2 * Zn + Fe + 6 * C + 6 * N = 2 * 65.4 + 52 + 6 * 12 + 6 * 14 = 2 * 65.4 + 66 + 84 = 270.8$ g which is Zn part.

- Zn part in $Zn_2Fe(CN)_6$: $in\ Zn_2Fe(CN)_6 = 2 * 65.4 / 342.6 * 100 \sim = 37.8 \%$.

- Atomic weight Zn / Molecular weight $Zn_2Fe(CN)_6 * 100 \sim = 37.8 / 100 * 100 \sim 10.5 \%$.

</think>

<answer>

A. 10 % Zn

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